Project Overview

Client and Context Background

Our story begins at **QiYao Educational Technology Company** in China. The client, **Ms. Zhao**, serves as the Director of English Language. To meet market demand, QiYao rapidly launched the **Think program for Grades 2–5**. The program utilizes Cambridge's *THINK* textbook series as its primary instructional source. At the time of analysis, the program served **67 students** across five distinct classes.

Because the program was launched at high speed to capture market share, several institutional foundations were omitted during initial rollout. No formal placement assessment was administered at enrollment; all classes simply began at *Think Starter Level*. Furthermore, explicit class-level learning outcomes and a coordinated assessment framework were not established. Teachers were left to follow textbook unit sequences while independently arranging practice, homework, and tests according to individual class progress.

This operational environment created a complex web of **stakeholder priorities**. Parents primarily evaluate course success through improved school examination performance, expressing concern over slow progress. Ms. Zhao recognizes this commercial risk—unaddressed parent concerns jeopardize student retention, word-of-mouth reputation, enrollment revenue, and QiYao's long-term vision of building a continuous K–12 learning pathway.

The Problem

Current Performance: Students demonstrate severe difficulty transferring isolated grammar knowledge into independent text production. Observed written work exhibits persistent tense errors, incorrect verb forms, missing auxiliary verbs (especially in negative constructions), incorrect word order, incomplete sentences in reading comprehension answers, and paragraph compositions plagued by off-topic content and sentence fragments.

Desired Performance: Learners must independently apply level-appropriate grammar to construct grammatically complete, contextually accurate sentences and connect them into coherent, prompt-relevant written texts without relying on continuous teacher scaffolding.

The Performance Gap A stark contrast emerged between observed learner output and desired program outcomes. While teachers reported noticeable improvements in students' oral communication and reading confidence, learners stumbled significantly in written production.

Instructional Root Causes The analysis revealed five major instructional breakdown points:

1. **Unclear Performance Expectations:** Program outcomes were never explicitly defined per level, making it difficult for teachers to benchmark independent mastery versus supported practice.
2. **Limited Transfer Opportunities:** Classroom activities heavily emphasized isolated single-structure grammar drills. Learners were rarely asked to select and combine multiple grammatical structures in authentic writing contexts.
3. **Insufficient Scaffolding for Written Production:** Textbook explanations were brief, and exercises jumped directly from mechanical grammar selection to open-ended writing prompts without step-by-step construction scaffolding.
4. **Inconsistent Feedback Loops:** Written assignment corrections lacked consistent detail. Teachers rarely provided structured opportunities for students to revise work or demonstrate understanding through follow-up application tasks.
5. **Prerequisite Skill Gaps:** Key sentence-level foundations (subject-verb agreement, sentence boundaries, pronoun references) were not systematically verified before introducing complex target structures.

Non-Instructional Environmental Constraints

Instruction alone cannot fix systemic organizational constraints. The analysis identified critical non-instructional factors influencing student output:

1. Heavy primary school homework demands, resulting in low completion of optional supplementary writing assignments.
2. Parental requests to limit additional weekend homework loads.
3. High teacher turnover and uneven teacher onboarding schedules.
4. Strict classroom time limits (two-hour weekly sessions) restricting live writing practice.

The Empirical Evidence:

1. **Student Assignment Artifacts:** Starter-level samples revealed pervasive tense errors, missing auxiliaries, and responses under 5 sentences that omitted prompt questions. Level 1 samples showed paragraph production capability but frequent sentence fragments, mixed-tense confusion, and off-topic drift.
2. **Teacher Qualitative Inputs:** Questionnaire insights provided crucial context. **Teacher A** noted that age does not equal proficiency (younger Grade 2–3 students often outpaced older classmates orally). **Teacher B** highlighted persistent structural traps (e.g., learners keeping the third-person *-s* suffix after auxiliary verbs *does/doesn't*). **Teacher C** observed a cognitive separation between idea generation and English expression, noting students could explain complex ideas in Chinese but struggled to translate them into structured English sentences.
3. **Publisher Pre-Test Statistics (67 Students):**

Think Pre-Test Placement Band	Score Range	Students	Percentage
A2	16–33	45	67.2%
B1	34–51	20	29.9%
A1	0–15	1	1.5%
B1+	52–69	1	1.5%
B2	70–87	0	0%
C1	88–112	0	0%

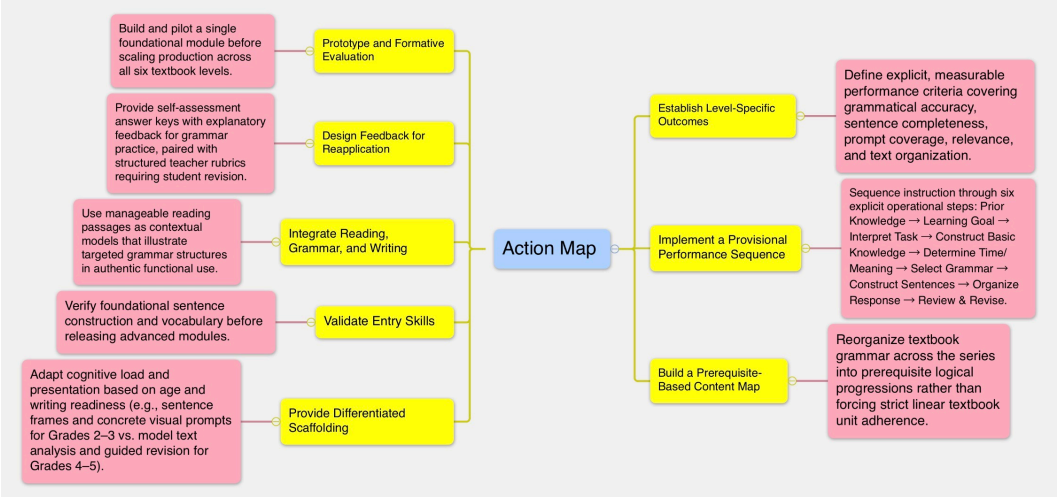
4. **Curriculum Unit Deep-Dive (Starter Unit 1):** Analysis of Starter Unit 1 showed targeted grammar (*be* verbs, question words) presented via brief rules and selected-response exercises, followed immediately by a personal writing prompt without any connecting written models or explicit sentence-building sequence.

The Solution Architecture

After gathering preliminary data analysis on the learners and consulting the stakeholders, I recommended an Supplementary Self-Assessment eLearning solution for several reasons:

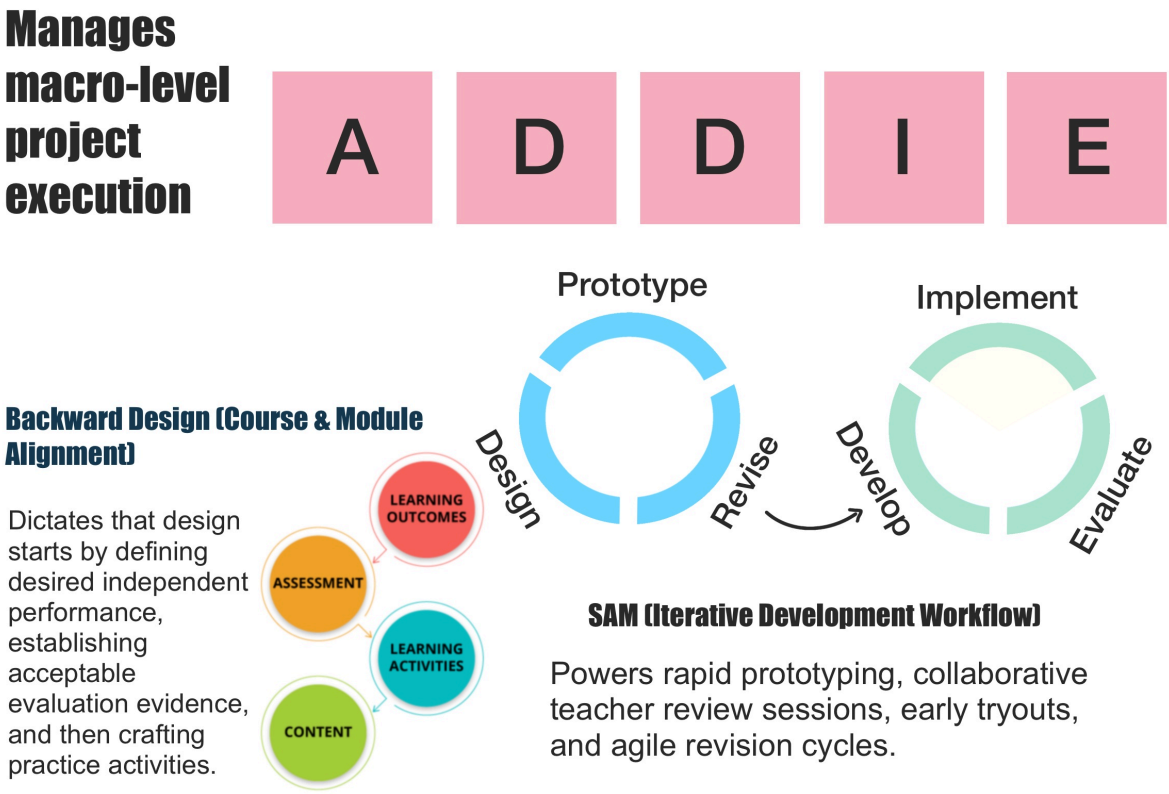
1. It address the performance gap meanwhile allows students to work at their own pace outside schedule classes without altering existing classroom textbook curriculum or school schedules.
2. It also overcome the lack of scaffolding or instructional sequences that connect grammar practice directly to independent written production.
3. Additionally, eLearning incorporates self-checking explanation alongside structured teacher feedback that require students to revise their written work, ensuring feedback leads directly to skill reapplication.
4. Lastly, class sessions are limited and school homework loads are heavy, therefore, this solution provides students more opportunities to practice.

Strategic Action Plan



Instructional Design Model

Rather than relying on a single design framework, the project integrates proven instructional models into a synchronized workflow:



Project Goals and Shared Course Outcomes

The overarching project goal is to build an age-responsive, self-paced supplementary e-learning product delivered on Notion that systematically transforms controlled grammar knowledge into independent written production.

All modules support six **Shared Course Outcomes (COs)**. After completing a learning sequence, students will independently:

- **CO1. Interpret Task Requirements:** Identify context, temporal relationships, and required details from prompts.
- **CO2. Apply Grammar in Context:** Select and correctly form target structures in single and mixed contexts.
- **CO3. Construct Complete Sentences:** Produce syntactically complete sentences with appropriate word order and necessary elements.
- **CO4. Produce Relevant Writing:** Organize sentences into coherent, prompt-focused written responses.
- **CO5. Review and Revise:** Diagnose and correct targeted grammatical, structural, and organizational errors.
- **CO6. Apply Learning Independently:** Execute target skills in novel contexts without teacher scaffolding.

Textbook Analysis & Curriculum Architecture

To eliminate learning gaps and address the textbook's distributed sequence, I conducted a full-series analysis of *THINK Starter through Level 5 with two teachers and the client*. This mapping consolidated repeated content and organized target language into **11 Core Content Strands** comprising **36 Curriculum Modules (M01–M36)**.

The 11 Full-Series Content Strands

- **Strand A: Sentence Foundations & Reference (M01–M04):** Pronouns, *be* verbs, questions/negatives, possession (*have got*), and existence (*there is/are*).

- **Strand B: Nouns, Articles, & Quantity (M05–M07):** Countable/uncountable nouns, articles (*a/an/the/zero*), quantifiers, and group reference.
- **Strand C: Present-Time Expression (M08–M10):** Present simple routines, present continuous temporary activities, and present simple vs. continuous contrasts.
- **Strand D: Past Events, Experiences, & Habits (M11–M16):** Past simple completed events, past continuous background actions, past habits (*sed to/would*), present perfect experiences/duration, and past perfect narration.
- **Strand E: Future Expression (M17–M18):** Intentions (*going to*), arrangements (present continuous), predictions (*will/might*), schedules, and future continuous/perfect.
- **Strand F: Modality (M19–M22):** Ability (*can/could/managed to*), permission/rules (*must/let/make*), obligation/advice (*should/have to/had better*), and deduction/past criticism (*must have/should have*).
- **Strand G: Interaction & Conditionals (M23–M26):** Question tags, agreement (*so/neither*), real conditionals (zero/first), hypothetical conditions (second/third), wishes, and mixed conditionals.
- **Strand H: Passive Voice & Reporting (M27–M30):** Passive tenses, causatives (*have something done*), reported speech (statements/questions/commands), and formal passive reporting/hedging.
- **Strand I: Verb Patterns & Expansion (M31–M33):** Gerunds vs. infinitives, phrasal verb object placement, relative clauses, and participle clauses.
- **Strand J: Modification & Emphasis (M34–M35):** Comparative/superlative modifiers, adverbials, cleft sentences, substitution/ellipsis, and inversion.
- **Strand K: Connections Between Ideas (M36):** Purpose (*in order to*), result (*so/such that*), contrast (*although/despite*), and cause/effect connectors.

Age-Differentiated Module Expectations

To accommodate cognitive maturity, each strand defines distinct performance expectations across two pathways:

- **Grades 2–3 Pathway:** Focuses on foundational sentence completeness, visual context prompts, short task sets, temporary sentence frames, oral rehearsal, and explicit guided checking.
- **Grades 4–5 Pathway:** Focuses on paragraph-level organization, contextual model analysis, multi-structure selection, independent planning, and self-directed rubric revision.

Subskill Functional Breakdown

Every module translates broad grammar headers into four operational subskill steps:

1. *Task & Meaning Interpretation:* Decoding prompt demands and temporal cues (CO1).
2. *Grammatical Selection:* Choosing correct structures for contextual meaning (CO2).
3. *Sentence Construction:* Synthesizing syntax, agreement, and vocabulary into complete sentences (CO3).
4. *Text Organization & Revision:* Assembling sentences into coherent outputs and self-correcting errors (CO4, CO5).

The 90-Module Learner-Facing Sequence

To make learning digestible, the 36 overarching curriculum containers were expanded into **90 provisionally defined learner-facing modules**, presented in a continuous sequential progression.

Prototype Development, Testing & Findings

With the curriculum architecture defined, the team built a functional digital prototype on **Notion** covering **Modules 1–15** ("Sentence Foundations and Personal Description") to conduct empirical tryouts.

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Module 1: I am...

Module 1: I am...

Level Start Level

Add a property

📍 Before you Start:

1. You know simple words for people, feelings, and ages. 你认识一些表示人物、感受和年龄的简单词汇。
2. You can tell who **I, you, he, she, it, we,** and **they** refer to. 你知道 I、you、he、she、it、we、they 分别指谁或什么。

🎯 Learning Goal

By the end of this module, you can:

1. Find pronoun and be verb in a sentence.
2. Write complete sentences about yourself.
3. Choose the right word: am, is, or are.

❓ Step 1: Is Something Wrong?

is a student.	I am from.	We happy.
She nine years old.	They friends.	It is a.

📒 Step 2: Learn The Basics

We use am, is, or are + a noun phrase to say who someone is or what something is.
我们用 am、is 或 are 加名词短语，说明某人是谁或某物是什么。

Subject Pronouns	Be	Noun Phrases (Singular)
I	am	a student
You	are	a student
He	is	a student
She	is	a student
It	is	a country
Subject Pronouns	**Be**	**Noun (Plural)**
We	are	students
You	are	students
They	are	students

We use am, is, or are + an adjective to describe someone or something.
我们用 am、is 或 are 加形容词，描述某人或某物。

Subject Pronouns	Be	Adjectives
I	am	happy
You	are	happy
He	is	tired
She	is	happy
It	is	big
We	are	tired
You	are	tired
They	are	happy

🔎 Step 3: Read & Find

A. Find the pronoun and the be verb.
找出每句中的代词和be动词。

Sentence	Subject Pronoun	Be Verb
1. I am happy.	I	am
2. He is tired.		
3. It is big.		
4. You are a student.		
5. She is nine years old.		
6. We are good friends.		
7. They are happy students.		
8. I am a happy student.		
9. He is a kind and happy boy.		
10. You are a kind and helpful friend.		

▾ Feedback

Item	Subject Pronoun(s)	Be Verb(s)
1	I	am
2	He	is
3	It	is
4	You	are
5	She	is
6	We	are
7	They	are
8	I	am
9	He	is
10	You	are

B. Read & Find
Read the paragraph. Mark the subject pronouns in blue and the be verbs in yellow.
阅读短文，用蓝色标记主语代词，用黄色标记be动词。

> I am Ben. I am nine years old. I am a student. I am happy today. This is my friend, Anna. She is nine years old. She is kind and helpful. We are good friends. We are happy students.

💙 subject pronoun 💛 be verb

▾ Feedback

> I am Ben. I am nine years old. I am a student. I am happy today. This is my friend, Anna. She is nine years old. She is kind and helpful. We are good friends. We are happy students.

🧠 Step 4: Think About the Meaning

Match each sentence with its meaning.
将每个句子与它表达的信息配对。

🧩 Step 5: Build a Sentence

A. Make a Sentence

Use the words. Add am, is, or are. Write a complete sentence.
使用提示词，加入 am、is 或 are，写出完整句子。

Information	Your sentence
I / a student	
She / happy	
He / nine years old	
You / a good friend	
It / big	
We / students	
They / tired	
I / happy today	

▾ Feedback:

Information	Your sentence
I / a student	I'm a student.
She / happy	She is happy.
He / nine years old	He is nine years old.
You / a good friend	You are a good friend.
It / big	It is big.
We / students	We are students.
They / tired	They are tired.
I / happy today	I am happy today.

✏️ Step 6: Write Your Own Work

Write 4–5 sentences about yourself. Use "I am" in each sentence.
写 4–5 句话介绍自己。每句话使用 I am。

Tell us:

- Your name
- Your age
- Who you are
- How you feel today?

My Writing:

✅ Step 7: Check Your Work

- ☐ I wrote about all four point. 我写到了以上四点。
- ☐ I used **I am** in each sentence. 每句话都有 I am。
- ☐ My sentences make sense. 我的句子意思完整。
- ☐ Each sentence starts with a capital letter. 每句话以大写字母开头。
- ☐ Each sentence ends with a period (.). 每句话以句号结尾。

The Two-Week Pilot Test

- *Pre-Pilot (June 8–12):* Confirm learner access/permissions, issue non-identifying learner codes, set up digital workspaces, and brief participating teachers.
- *Week 1 (June 15–19):* Learners work through Modules 1–7 independently at their own pace.
- *Week 2 (June 22–26):* Learners complete Modules 8–14 and attempt the Module 15 capstone.
- *Post-Pilot Assessment (July 3):* Administer aligned post-use learning assessment and Module 15 integrated writing task.
- *Data Scoring & Analysis (July 6):* Score student work artifacts using standardized rubric criteria.

Formative Assessment Findings:

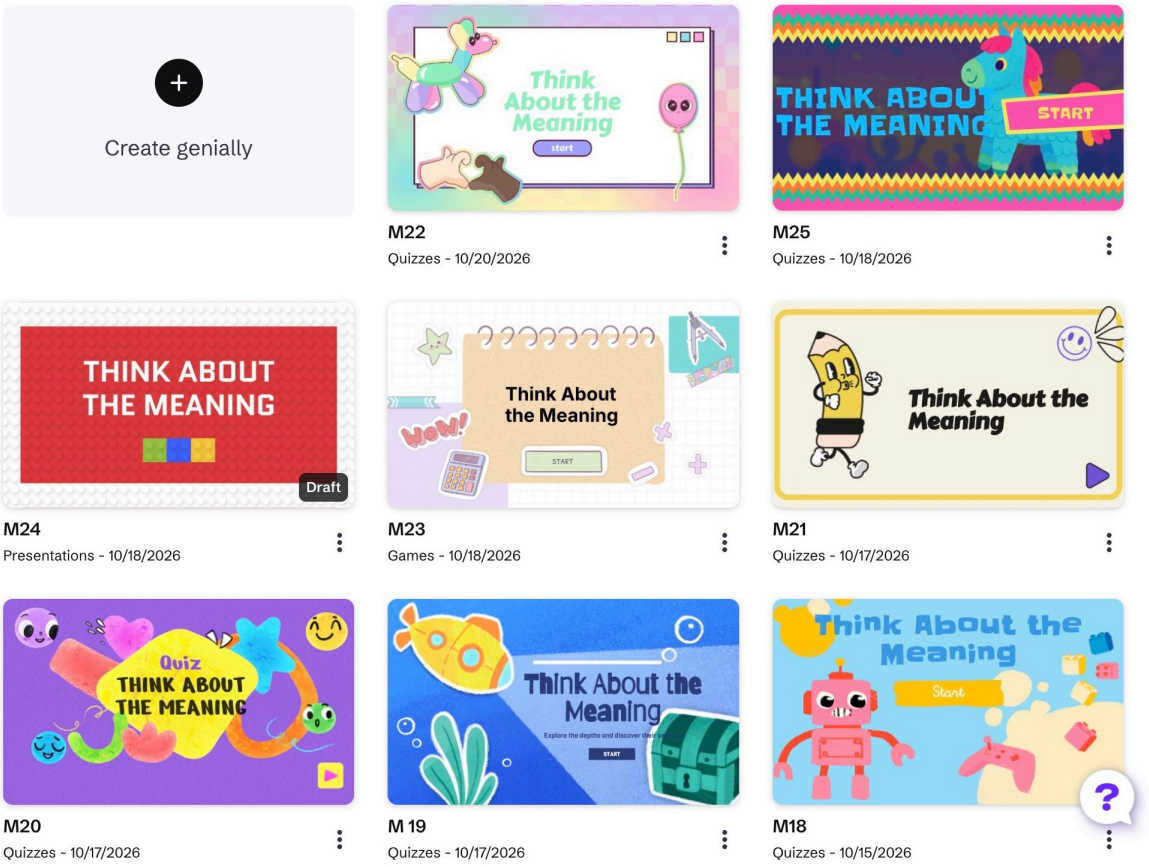
Student performance data collected post-pilot demonstrated compelling instructional progress:

- **Tasks 1–4 Performance:** Students scored remarkably high on foundational items, showing significant score gains. (Designers noted prior textbook exposure likely contributed to high baseline familiarity).
- **Task 5 Sentence Building:** Teacher observations recorded dramatic improvements in sentence construction speed, syntactic correctness, and student confidence during sentence-building tasks.
- **Module 15 Integrated Writing Task:** Final writing artifacts contained a **higher volume of complete sentences** and a **marked reduction in structural grammatical errors** compared to baseline assignments.
- *Assessment Revisions:* The team resolved to retain the core sentence-building mechanics while developing parallel evaluation items using unstudied prompt contexts to rigorously test far-transfer capabilities.

Formative Evaluation Findings:

Product quality, technical accessibility, and user experience evaluation yielded critical design insights:

- **Notion Usability Disparity:** While Grades 4–5 learners navigated Notion seamlessly and responded enthusiastically to the digital interface, **Grades 2–3 learners experienced severe navigation friction**, struggling with digital page structures.
    - ***Decision**:* Develop a high-quality **printable paper workbook option** for Grades 2–3 that mirrors digital objectives and scaffolding, while retaining Notion for Grades 4–5.
- **Bilingual Scaffolding Needs:** Learners across both pathways spent excessive cognitive energy attempting to translate navigation directions and self-review checklists.
    - ***Decision**:* Introduce **concise Chinese bilingual translations** specifically for learning objectives, task instructions, and self-checking checklists, maintaining English strictly for core instructional content and target language practice.
- **Cognitive Load Trap in Step 4 ("Think About the Meaning"):** Step 4 was designed to check conceptual comprehension. However, it utilized standard multiple-choice questions (MCQs). Because young learners were unfamiliar with test format mechanics, the MCQs created severe cognitive overload, distracting from the actual language concept.
    - ***Decision**:* Completely **redesign Step 4 from multiple-choice testing into a concrete, interactive visual sorting activity**, eliminating unnecessary reading traps while testing conceptual understanding.



Developing Interactive Activities in Genially

Full Development & AI-Generated Visual Mockups

After collecting and applying formative feedback from the prototype, I developed the remaining 75 modules. I gathered feedback from other teachers, learners, and my client to have a high-quality final product.

Roles and Responsibilities

Role	Responsibility
Instructional designer / project lead	Maintain alignment, produce content and tools, assemble the prototype, analyze evidence, and document revisions.

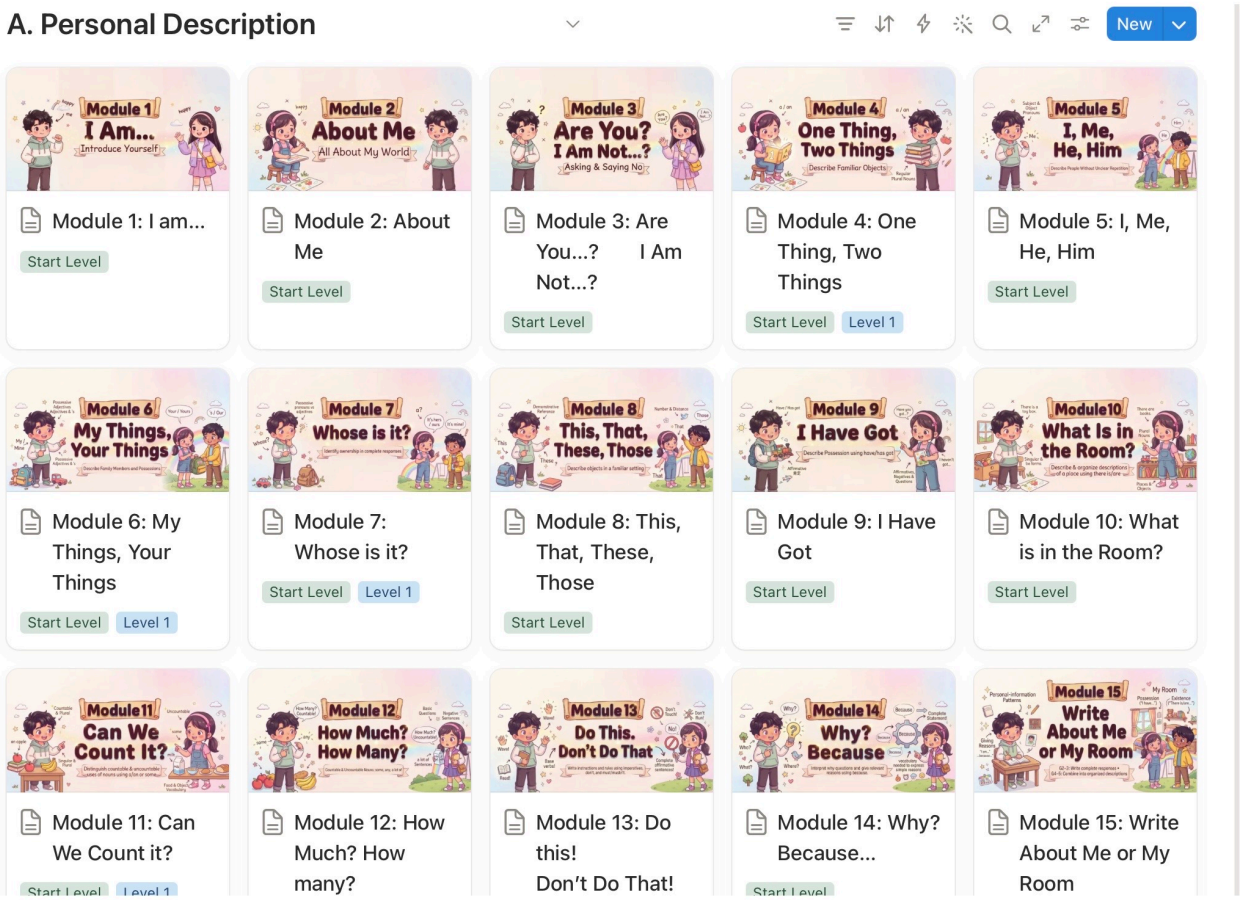
Client / program representative	Confirm priorities, constraints, platform and access decisions, and acceptable implementation conditions.
Participating teacher(s)	Review content and age appropriateness, judge task realism, help interpret learner errors, and collaborate with the designer
Learners (if an approved tryout occurs)	Attempt the activities and provide age-appropriate feedback; participation is subject to applicable permission and consent procedures.
Technical/platform support (if available)	Confirm access, permissions, compatibility, and practical response collection.

AI-Generated Visual Mockups

To establish a clear visual identity for young learners, I developed and refined prompts to guide the AI in generating visual covers that reflects the module title and learning goal.

AI-Generated Cover (1st Version)

AI-Generated Cover (2nd Version)



Final Version